IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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GitHub repository: https://github.com/darizek-art/ibm-data-science-capstone

Outline

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Executive Summary

A data-driven assessment of Falcon 9 first-stage recovery.

Historic data show a 66.7% landing-success rate in the predictive dataset.

Three classifiers tied for the highest test accuracy: 83.3%.

Introduction

Can Falcon 9 first-stage landing success be predicted from launch characteristics?

Successful recovery matters because reused first stages can reduce launch cost.

Data Collection and Methodology

Methodology

Methodology

- Collected launch, payload, booster, and launch-site data from SpaceX sources and Wikipedia tables.
- Cleaned records, created the binary landing-success label, and prepared model features.
- Used visual and SQL analysis to identify relationships among payload, orbit, site, and outcomes.
- Built and evaluated classification models using a held-out test set.

Data Collection

Two complementary sources were used:

- SpaceX REST data for launch, rocket, payload, core, and launchpad records
- Wikipedia launch tables for historic mission and landing information

The data sources were transformed into analysis-ready CSV files.

Data Collection – SpaceX API

REST calls retrieved launch records, then related endpoint data supplied rocket, payload, core, and launchpad details.

The final API dataset contains the launch variables needed for subsequent wrangling and modelling.

Data Collection – Web Scraping

Wikipedia tables were read with pandas, cleaned, and converted into structured columns.

Web scraping provided a broader historic launch view and supported the SQL and dashboard exercises.

Data Wrangling

- Retained single-core, single-payload Falcon 9 records.
- Converted dates and removed unsuitable records.
- Created the Class label: 1 for a successful landing and 0 otherwise.
- Result: 90 model-ready observations; 60 successful and 30 unsuccessful landings.

EDA with Data Visualization

Scatter plots compared payload, flight number, launch site, and orbit type.

- Bar charts measured orbit-level landing success, while a line chart showed the yearly trend.

- These views exposed important feature relationships before modelling.

EDA with SQL

SQL queries were used to:

identify launch sites and CCA records

- calculate payload totals and averages
- find first successful ground-pad landings
- compare mission and landing outcomes
- examine historic drone-ship and maximum-payload launches

Interactive Map with Folium

- Built an interactive map with launch-site markers for CCAFS, KSC, and VAFB.
- Added colour-coded landing outcomes to compare success and failure patterns.
- Used proximity markers and distance labels to explore each launch site's surroundings.

Build a Dashboard with Plotly Dash

The dashboard combines a launch-site selector, a payload range slider, a success pie chart, and a payload-versus-outcome scatter plot.

These controls make launch outcomes easier to explore by site, payload, and booster category.

Predictive Analysis (Classification)

- Standardised features and split the data into training and test sets.
- Evaluated Logistic Regression, SVM, Decision Tree, and KNN classifiers.
- Compared test accuracy and confusion matrices to select the strongest result.

Results

Key results

- Landing success in the modelling dataset: 66.7%
- EDA showed outcome patterns by orbit, site, payload, and time
    - Interactive tools supported geographic and launch-site exploration
    - Best test accuracy: 83.3%

Exploratory Data Analysis

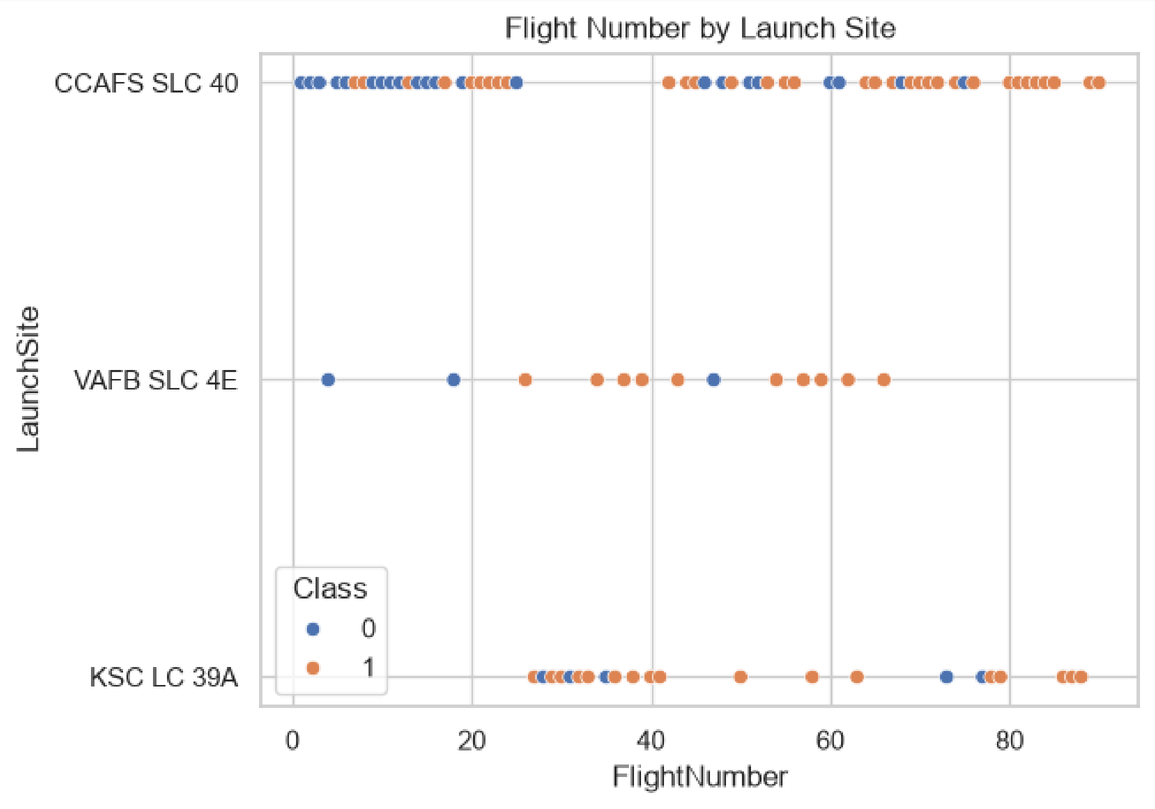
Insights drawn from EDA

Evidence-based analysis of flight number, payload mass, launch site, orbit type, and yearly landing-success trends.

Flight Number vs. Launch Site

Flight number increases over time for every site, reflecting the growth of Falcon 9 operations.

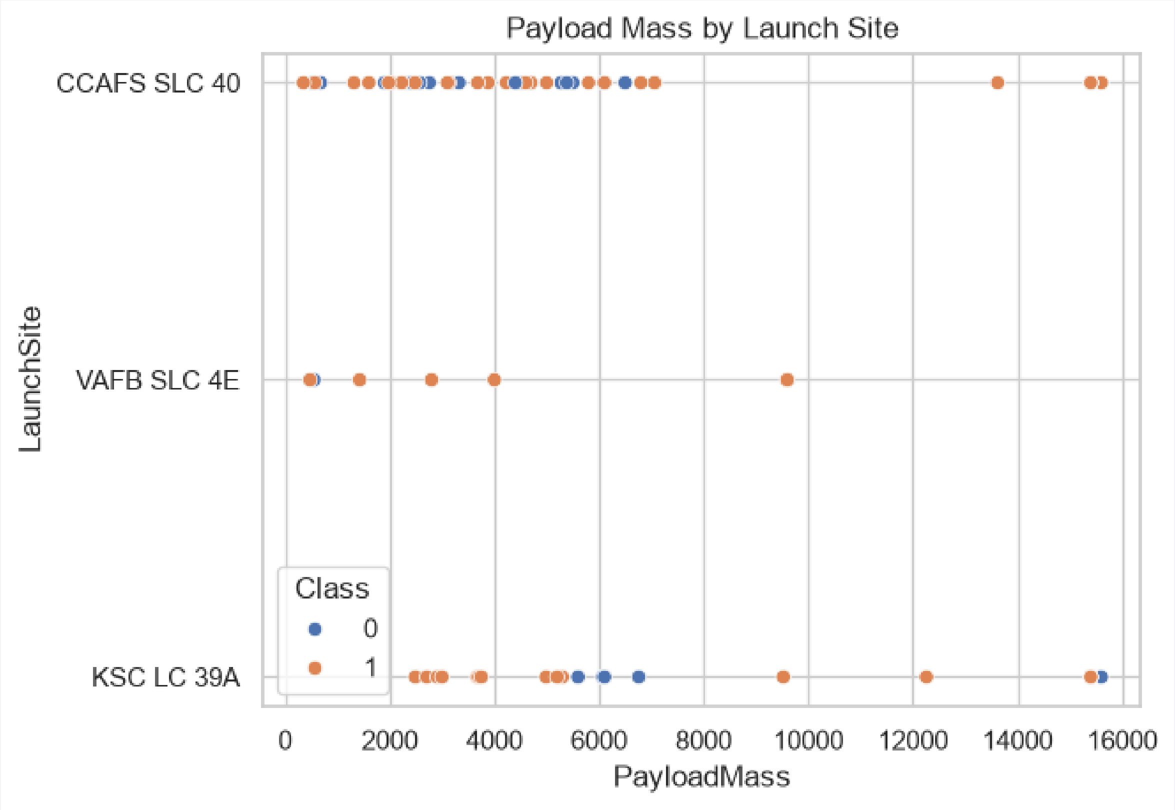
CCAFS SLC-40 has the broadest historic range of flight numbers; KSC LC-39A appears later in the sequence.



Payload vs. Launch Site

Payload mass varies substantially within and across launch sites.

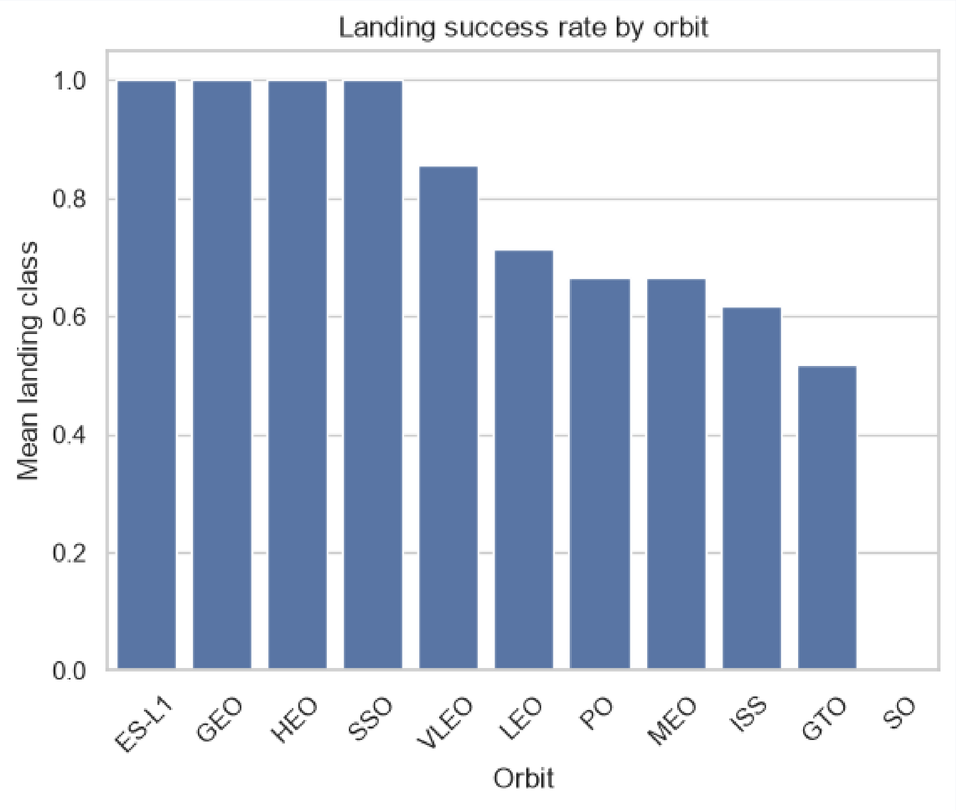
Later launches include more high-mass missions, so payload should be retained as a predictive feature.



Success Rate vs. Orbit Type

Orbit type is strongly associated with landing outcome.

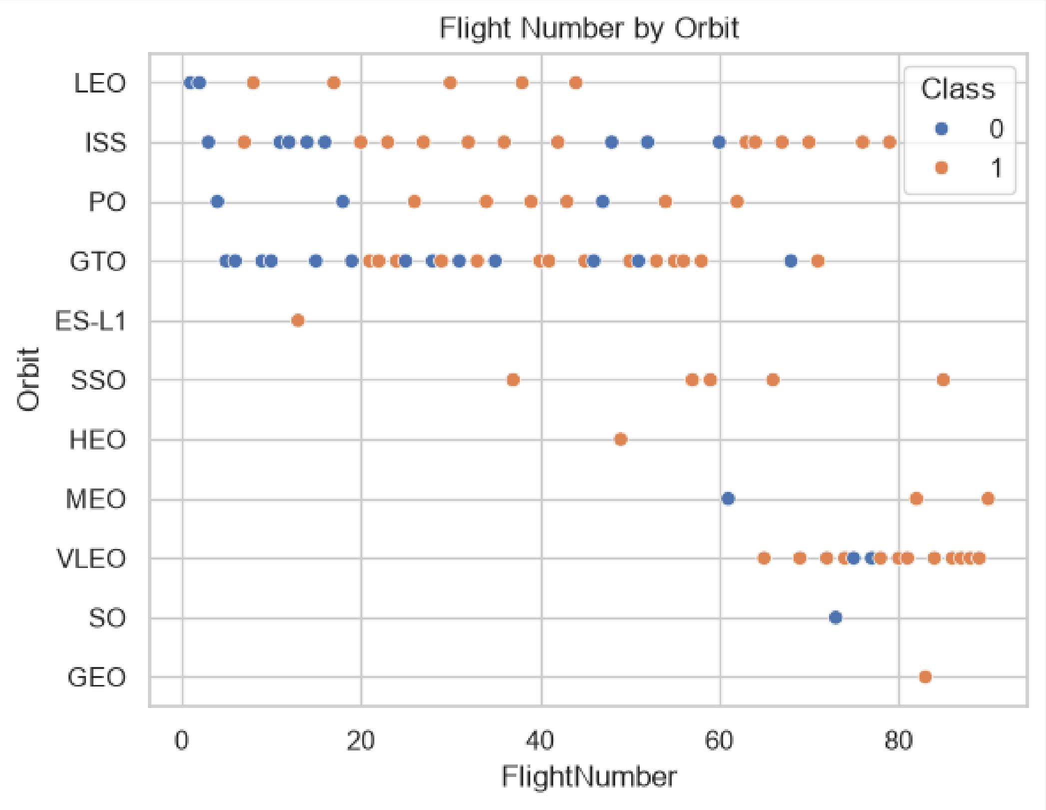
Several low-volume orbit categories show perfect success rates, while others have mixed or low success; results should be interpreted with sample size in mind.



Flight Number vs. Orbit Type

Later flight numbers cover a wider set of orbits.

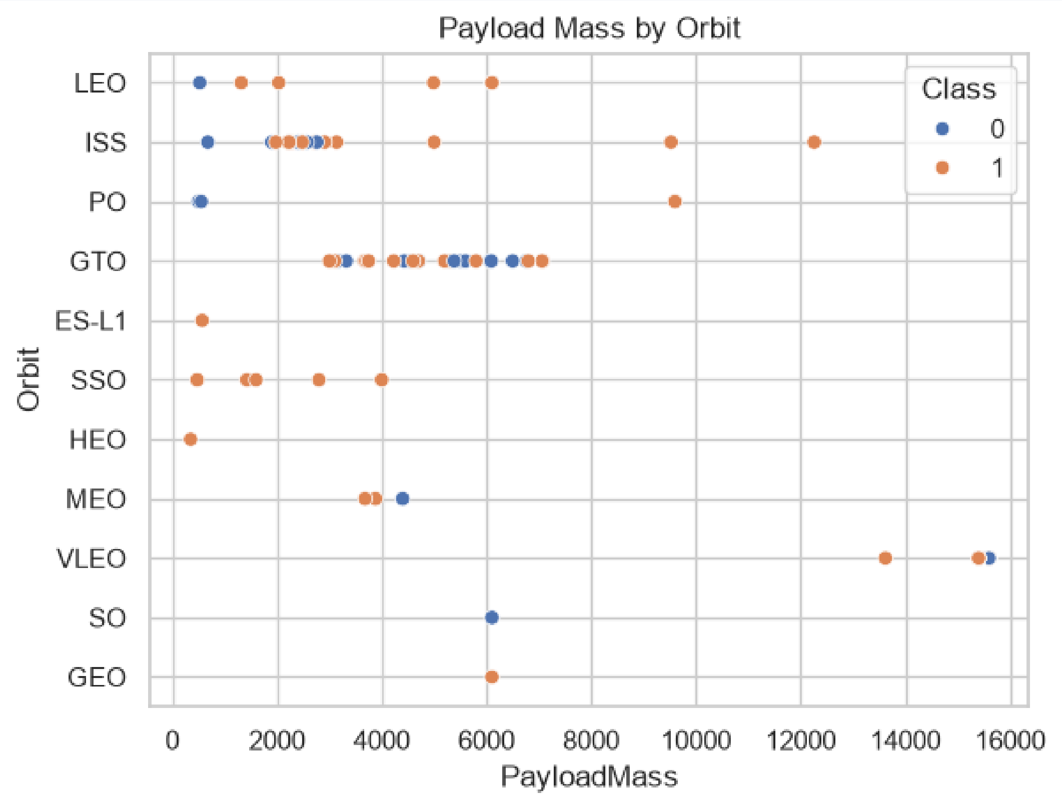
This suggests that operational maturity and mission profile both changed over time.



Payload vs. Orbit Type

Payload values overlap across orbit types, but some orbit categories concentrate in distinct mass ranges.

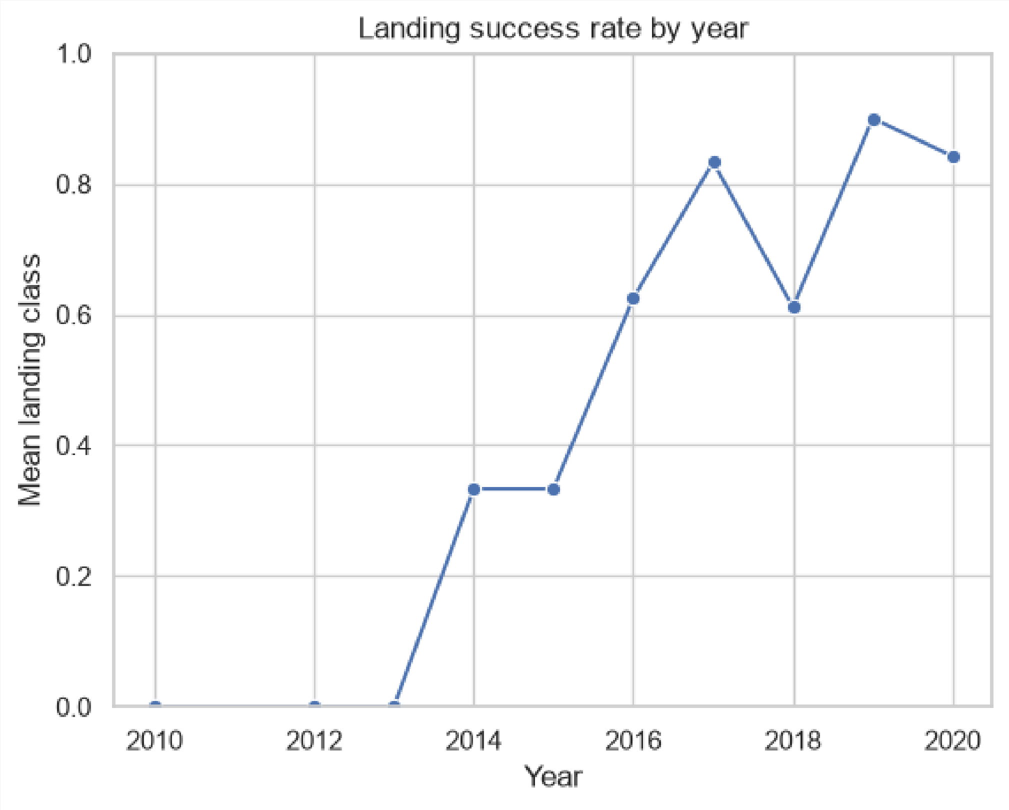
Payload and orbit therefore provide complementary information for classification.



Launch Success Yearly Trend

The yearly average landing-success rate trends upward as recovery operations mature.

The pattern supports including year and flight number as predictive features.



All Launch Site Names

Four unique sites were returned: CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, and VAFB SLC-4E.

These locations form the main geographic and operational comparison groups.

Launch Site Names Begin with 'CCA'

The first five CCA records are early CCAFS LC-40 launches, from B0003 through B0007.

They include the 2010 Falcon 9 debut and early Dragon demonstration missions.

Total Payload Mass

NASA Commercial Resupply Services payload mass total: 48,213 kg.

This query aggregates payload mass for Customer values containing NASA (CRS).

Average Payload Mass by F9 v1.1

Average payload mass for F9 v1.1 missions: 2,534.67 kg.

The value gives a useful baseline for the earlier Falcon 9 configuration.

First Successful Ground Landing Date

First successful ground-pad landing: 22 December 2015.

This marks an important milestone in first-stage recovery history.

Successful Drone Ship Landing with Payload between 4000 and 6000

Successful drone-ship recoveries between 4,000 and 6,000 kg included F9 FT B1026 (4,600 kg), B1022 (4,696 kg), B1031.2 (5,200 kg), and B1021.2 (5,300 kg).

Total Number of Successful and Failure Mission Outcomes

Mission outcomes were overwhelmingly successful: 98 standard successes, one success with unclear payload status, one additional success entry, and one in-flight failure.

Boosters Carried Maximum Payload

Maximum payload mass: 15,600 kg.

Several Block 5 boosters reached this value, demonstrating the capacity of later Falcon 9 configurations.

2015 Launch Records

2015 drone-ship landing failures occurred in January (F9 v1.1 B1012) and April (F9 v1.1 B1015), both from CCAFS LC-40.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

From 2010-06-04 to 2017-03-20, 'No attempt' was most common (10). Drone-ship successes and failures were tied at five each, illustrating the learning curve of recovery operations.

Interactive Visual Analytics

Launch Sites Proximities Analysis

The Folium map and dashboard turn the launch data into explorable geographic and operational views.

Launch Sites on the Global Map

Folium map result:

Four launch-site markers identify the principal locations at Cape Canaveral, Kennedy Space Center, and Vandenberg. Interactive pop-ups provide site details and support geographic comparison.

Landing Outcomes by Launch Site

Folium map result:

Colour-coded markers distinguish successful and unsuccessful landing attempts. This view makes outcome patterns by launch site visible without relying only on a table.

Launch-Site Proximity Analysis

Folium map result:

The selected-site view layers coastline, highway, and railway features with distance labels. These map objects provide contextual evidence about launch-site surroundings.

Plotly Dash Dashboard

Build a Dashboard with Plotly Dash

Dashboard evidence: launch-site outcome comparison, payload filtering, and booster-version patterns.

Dash: Launch Outcomes for All Sites

Dashboard result:

The all-sites pie chart compares successful launches by site. The launch-site dropdown updates the chart so the user can move from programme-level totals to site-specific outcomes.

Dash: Launch-Site Success Comparison

Dashboard result:

Selecting one launch site changes the pie chart to successful versus unsuccessful launches at that site. This interaction makes the site-level success ratio directly comparable.

Dash: Payload, Booster Version, and Outcome

Dashboard result:

The payload-range slider filters the scatter plot, while colour encodes booster version. Together, the controls expose relationships among payload, hardware, launch site, and landing success.

Predictive Analysis

Predictive Analysis (Classification)

Classification results compare four models and identify the strongest observed test performance

Classification Accuracy

Test accuracy by model:

- Logistic Regression — 83.3%
- Support Vector Machine — 83.3%
- Decision Tree — 77.8%
- K-Nearest Neighbours — 83.3%

Three models tied for the highest observed test accuracy.

Confusion Matrix

The best-performing models were tied at 83.3% test accuracy.

The confusion-matrix review is used to compare correct and incorrect landing predictions, with particular attention to the small held-out sample.

Conclusions

- Falcon 9 landing success rose as operations matured.
- Launch site, orbit, payload, and flight number are informative features.
- Historic landing data supported useful classification performance.
- The 83.3% top test score should be interpreted cautiously because the dataset is small and the best models were tied.

Appendix

Project artefacts include:

- API and web-scraping notebooks
- cleaned datasets for wrangling and EDA
- SQL, Folium, and Dash analyses
- classification-model notebook and results

All artefacts are organised in the project repository.